Swinburne University of Technology

Linear Algebra and Applications

TUTORIAL 4: Rank and nullity of a matrix, vectors and vector spaces

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- 1. Determine the rank and the nullity of the following matrices

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -4 & -2 \\ 3 & 6 & 3 \end{bmatrix}.$$

- 2. Let A be a 4×3 matrix that has a full rank.

  (a) Solve $AX = 0$

  (b) How many free parameters are in the set of all solutions of $A^T X = 0$?

- 3. The vertices of a triangle are located in the following points $A = (1, 2, 3)$, $B = (-1, 5, -7)$ and $C = (2, 0, 4)$.

  (a) Give a vector representation of the median that joins vertex A with the opposite side.

  (b) Calculate the length l of the median in part (a).

- 4. The position vectors of points P and Q are $2\boldsymbol{i} - 3\boldsymbol{j} + 4\boldsymbol{k}$ and $3\boldsymbol{i} - 7\boldsymbol{j} + 12\boldsymbol{k}$, respectively. Determine the unit vector in the direction of $\overline{PQ}$. What are the direction cosines of $\overline{PQ}$?

- 5. Determine if the vector $\boldsymbol{v} = \boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k}$ can be represented as a linear combination of vectors

  (a) $\boldsymbol{u}_1 = \boldsymbol{i} + \boldsymbol{j}$, $\boldsymbol{u}_2 = 2\boldsymbol{i} + 3\boldsymbol{j} + 4\boldsymbol{k}$ and $\boldsymbol{u}_3 = 3\boldsymbol{i} + 8\boldsymbol{j} + 20\boldsymbol{k}$

  (b) $\boldsymbol{u}_1 = \boldsymbol{i} + \boldsymbol{j} - 3\boldsymbol{k}$ and $\boldsymbol{u}_2 = \boldsymbol{k}$

- 6. Show that the set of vectors

$$\boldsymbol{u} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

  such that $x + y + z = 0$, is a vector space over the real numbers.

Solutions

- 1. Using Gaussian elimination we reduce A and B to row echelon form

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \underset{R_2 \to R_2 + R_1}{\Rightarrow} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix} \underset{R_3 \to 2R_3 - R_2}{\Rightarrow} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Consequently

$$\mathrm{Rank}(A) = 2.$$

Similarly

$$B = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -4 & -2 \\ 3 & 6 & 3 \end{bmatrix} \underset{\substack{R_2 \to R_2 + 2R_1 \\ R_3 \to R_3 - 3R_1}}{\Rightarrow} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

so that

$$\mathrm{Rank}(B) = 1.$$

Now we need to solve a homogeneous system of equations $AX = 0$, or, explicitly

$$A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Using A in REF, we find

$$\begin{array}{ll} x_3 & \text{is free} \\ x_2 & +x_3 = 0, \ \to x_2 = -x_3 \\ x_1 & +x_2 + x_3 = 0, \ \to x_1 = -x_2 - x_3 = -(-x_3) - x_3 = 0. \end{array}$$

Solution in the vector form is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \quad x_3 \in (-\infty, +\infty).$$

This solution contain only one free parameter, therefore, the Nullity of A is

$$\mathrm{Nullity}(A) = 1.$$

Note that

$$\mathrm{Nullity}(A) + \mathrm{Rank}(A) = 2 + 1 = 3$$

as it should be.

Similarly, for the matrix B we set $BX = 0$ and find that x_3 and x_2 are free and

$$x_1 = -2x_2 - x_3.$$

Or in the vector form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Two free parameters shows that

$$\mathrm{Nullity}(A) = 2.$$

- 2. Since A is a 4×3 matrix that has full rank, we conclude that

$$\mathrm{Rank}(A) = \min(4, 3) = 3.$$

(a) Using the Rank-Nullity theorem

$$\text{Nullity}(A) + \text{Rank}(A) = n,$$

where $n = 3$ is the number of columns of A, we conclude that

$$\text{Nullity}(A) + 3 = 3 \to \text{Nullity}(A) = 0.$$

Therefore, the set of solutions of the homogeneous system $AX = 0$ has no free parameters. This can only be true for the trivial solution, i.e. $X = 0$.

(b) The transposed matrix A^T is an 3×4 matrix with rank

$$\text{Rank}(A^T) = \text{Rank}(A) = 3.$$

From the Rank-Nullity theorem, we obtain

$$\text{Nullity}(A^T) + \text{Rank}(A^T) = \text{Nullity}(A^T) + 3 = 4.$$

Therefore,

$$\text{Nullity}(A^T) = 1.$$

This shows that the set of solutions to $A^T X = 0$ has one free parameter.

- 3. (a) In the triangle ABC, the side BC can be represented as a vector

$$\overline{BC} = (2 - (-1))\boldsymbol{i} + (0 - 5)\boldsymbol{j} + (4 - (-7))\boldsymbol{k} = 3\boldsymbol{i} - 5\boldsymbol{j} + 11\boldsymbol{k}.$$

Let D be the middle point of the side BC. Then the vector $\overline{BD}$ is

$$\overline{BD} = \frac{1}{2}\overline{BC} = \frac{3}{2}\boldsymbol{i} - \frac{5}{2}\boldsymbol{j} + \frac{11}{2}\boldsymbol{k}.$$

The median $\overline{AD}$ can be expressed as

$$\overline{AD} = \overline{AB} + \overline{BD} = \overline{AB} + \frac{1}{2}\overline{BC}$$

Finally

$$\overline{AD} = (-1 - 1)\boldsymbol{i} + (5 - 2)\boldsymbol{j} + (-7 - 3)\boldsymbol{k} + \frac{3}{2}\boldsymbol{i} - \frac{5}{2}\boldsymbol{j} + \frac{11}{2}\boldsymbol{k} = -\frac{1}{2}\boldsymbol{i} + \frac{1}{2}\boldsymbol{j} - \frac{9}{2}\boldsymbol{k}.$$

Note also that the coordinates of D can be also calculated the mean of the corresponding coordinates of points B and C:

$$D = \left(\frac{-1+2}{2}, \frac{5+0}{2}, \frac{4-7}{2}\right) = \left(\frac{1}{2}, \frac{5}{2}, -\frac{3}{2}\right).$$

Then

$$\overline{AD} = \left(-\frac{1}{2}, \frac{1}{2}, -\frac{9}{2}\right)$$

(same as above).

(b) The length of the median AD is

$$AD = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{81}{4}} = \frac{1}{2}\sqrt{83}.$$

- 4. The vector $\overline{PQ}$ is

$$\overline{PQ} = (3-2)\boldsymbol{i} + (-7-(-3))\boldsymbol{j} + (12-4)\boldsymbol{k} = \boldsymbol{i} - 4\boldsymbol{j} + 8\boldsymbol{k}.$$

The unit vector $\boldsymbol{u}$ in the direction of $\overline{PQ}$ is

$$\boldsymbol{u} = \frac{\overline{PQ}}{PQ} = \frac{1}{\sqrt{1+16+64}}(\boldsymbol{i} - 4\boldsymbol{j} + 8\boldsymbol{k}) = \frac{1}{9}\boldsymbol{i} - \frac{4}{9}\boldsymbol{j} + \frac{8}{9}\boldsymbol{k}.$$

The directions cosines of $\boldsymbol{u}$ coincide with the respective coordinates

$$\cos(\alpha) = \frac{1}{9},\ \cos(\beta) = -\frac{4}{9},\ \cos(\gamma) = \frac{8}{9}.$$

- 5. (a) We require that

$$\boldsymbol{v} = \boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k} = a\boldsymbol{u}_1 + b\boldsymbol{u}_2 + c\boldsymbol{u}_3 = a[\boldsymbol{i} + \boldsymbol{j}] + b[2\boldsymbol{i} + 3\boldsymbol{j} + 4\boldsymbol{k}] + c[3\boldsymbol{i} + 8\boldsymbol{j} + 20\boldsymbol{k}],$$

where a, b, c are to be found. Comparing the respective coordinates of vectors in the left- and the right-hand sides, we obtain

$$\begin{aligned} 1 &= a + 2b + 3c \\ 1 &= a + 3b + 8c \\ 1 &= 4b + 20c \end{aligned}$$

To solve this set of equations we use Gaussian elimination

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 1 & 3 & 8 & 1 \\ 0 & 4 & 20 & 1 \end{array}\right] \underset{R_2 \to R_2 - R_1}{\Rightarrow} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 5 & 0 \\ 0 & 4 & 20 & 1 \end{array}\right] \underset{R_3 \to R_3 - 4R_2}{\Rightarrow} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

This system is inconsistent and, therefore, the vector $\boldsymbol{v}$ cannot be represented as a linear combination of $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$.

(b) Similarly to part (a), we reqire

$$\boldsymbol{v} = \boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k} = a\boldsymbol{u}_1 + b\boldsymbol{u}_2 = a[\boldsymbol{i} + \boldsymbol{j} - 3\boldsymbol{k}] + b\boldsymbol{k}.$$

This implies

$$1 = a,\ \ 1 = a,\ \ 1 = b - 3a.$$

This system is consistent, the solution is

$$a = 1,\ \ b = 4.$$

Therefore, the vector $\boldsymbol{v}$ can be represented as

$$\boldsymbol{v} = \boldsymbol{u}_1 + 4\boldsymbol{u}_2.$$

- 6. For a set V to be a vector space over the real numbers, we require that for any two elements $\boldsymbol{v}$ and $\boldsymbol{u}$ from V and any scalar λ the sum $\boldsymbol{v} + \boldsymbol{u}$ and $\lambda\boldsymbol{v}$ are also in V. Thus we need to check that if

$$\boldsymbol{v} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix},\quad \boldsymbol{u} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix},$$

such that $x_1 + y_1 + z_1 = 0$ and $x_2 + y_2 + z_2 = 0$ then for

$$\boldsymbol{v} + \boldsymbol{u} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + x_2 \end{bmatrix},$$

the sum of the coordinates is also zero, i.e we check that

$$x_1 + x_2 + y_1 + y_2 + z_1 + z_2 = (x_1 + y_1 + z_1) + (x_2 + y_2 + z_2) = 0.$$

Similarly, we check that for

$$\lambda \boldsymbol{v} = \lambda \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda y_1 \\ \lambda z_1 \end{bmatrix},$$

the sum of the coordinates is zero, i.e.

$$\lambda x_1 + \lambda y_1 + \lambda z_1 = 0.$$

Thus, the set of vectors

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad x + y + z = 0$$

is a vector space over the real numbers.